

mftd Extended Colour Palette Guide

2026 firmware, colour map 1 (Expanded). Fixed 128-entry lookup, one byte per colour.
Device-wide setting at config address 33. Values exact, from the user guide's vector fills.
ENUM names are read from `mftd.constants.Color`, so this file cannot drift from the code.
Base-14 fonts only: intended to survive ``pdftotext -layout`` with columns intact.

1. Neutrals

7 constants, every achromatic byte in the palette

BYTE	HEX		R	G	B	ENUM	ALIAS
0	#000000		0	0	0	BLACK	
1	#1E1E1E		30	30	30	GREY_DARKEST	GRAY_DARKEST
71	#202020		32	32	32	GREY_DARK	GRAY_DARK
117	#404040		64	64	64	GREY_FULL	GRAY_FULL
118	#757575		117	117	117	GREY_LIGHT	GRAY_LIGHT
2	#7F7F7F		127	127	127	GREY_LIGHTTEST	GRAY_LIGHTTEST
3	#FFFFFF		255	255	255	WHITE	

2. Hue families

bytes 4-59, 14 families x 4 shades

BYTE	HEX		R	G	B	FAMILY	HUE	SHADE	ENUM
4	#FF4C4C		255	76	76	RED	0	LIGHT	RED_LIGHT
5	#FF0000		255	0	0	RED	0	FULL	RED_FULL
6	#7F0000		127	0	0	RED	0	DARK	RED_DARK
7	#1E0000		30	0	0	RED	0	DARKEST	RED_DARKEST
8	#FFBD6C		255	189	108	ORANGE	20	LIGHT	ORANGE_LIGHT
9	#FF5400		255	84	0	ORANGE	20	FULL	ORANGE_FULL
10	#591D00		89	29	0	ORANGE	20	DARK	ORANGE_DARK
11	#271B00		39	27	0	ORANGE	20	DARKEST	ORANGE_DARKEST
12	#FFFF4C		255	255	76	YELLOW	60	LIGHT	YELLOW_LIGHT
13	#FFFF00		255	255	0	YELLOW	60	FULL	YELLOW_FULL
14	#595900		89	89	0	YELLOW	60	DARK	YELLOW_DARK
15	#191900		25	25	0	YELLOW	60	DARKEST	YELLOW_DARKEST
16	#88FF4C		136	255	76	LIME	100	LIGHT	LIME_LIGHT
17	#54FF00		84	255	0	LIME	100	FULL	LIME_FULL
18	#1D5900		29	89	0	LIME	100	DARK	LIME_DARK
19	#142B00		20	43	0	LIME	100	DARKEST	LIME_DARKEST
20	#4CFF4C		76	255	76	GREEN	120	LIGHT	GREEN_LIGHT
21	#00FF00		0	255	0	GREEN	120	FULL	GREEN_FULL
22	#007F00		0	127	0	GREEN	120	DARK	GREEN_DARK
23	#001E00		0	30	0	GREEN	120	DARKEST	GREEN_DARKEST
24	#4CFF5E		76	255	94	EMERALD	126	LIGHT	EMERALD_LIGHT
25	#00FF19		0	255	25	EMERALD	126	FULL	EMERALD_FULL
26	#00590D		0	89	13	EMERALD	126	DARK	EMERALD_DARK
27	#001902		0	25	2	EMERALD	126	DARKEST	EMERALD_DARKEST
28	#4CFF88		76	255	136	JADE	133	LIGHT	JADE_LIGHT
29	#28F656		40	246	86	JADE	133	FULL	JADE_FULL
30	#0B681F		11	104	31	JADE	133	DARK	JADE_DARK
31	#002413		0	36	19	JADE	133	DARKEST	JADE_DARKEST
32	#4CFFB7		76	255	183	MINT	156	LIGHT	MINT_LIGHT
33	#00FF99		0	255	153	MINT	156	FULL	MINT_FULL
34	#005935		0	89	53	MINT	156	DARK	MINT_DARK
35	#001912		0	25	18	MINT	156	DARKEST	MINT_DARKEST
36	#4CC3FF		76	195	255	AZURE	200	LIGHT	AZURE_LIGHT
37	#00A9FF		0	169	255	AZURE	200	FULL	AZURE_FULL
38	#004152		0	65	82	AZURE	200	DARK	AZURE_DARK
39	#001019		0	16	25	AZURE	200	DARKEST	AZURE_DARKEST
40	#4C88FF		76	136	255	COBALT	220	LIGHT	COBALT_LIGHT
41	#0055FF		0	85	255	COBALT	220	FULL	COBALT_FULL
42	#001D59		0	29	89	COBALT	220	DARK	COBALT_DARK
43	#000819		0	8	25	COBALT	220	DARKEST	COBALT_DARKEST
44	#4C4CFF		76	76	255	BLUE	240	LIGHT	BLUE_LIGHT
45	#0000FF		0	0	255	BLUE	240	FULL	BLUE_FULL
46	#00007F		0	0	127	BLUE	240	DARK	BLUE_DARK
47	#00001E		0	0	30	BLUE	240	DARKEST	BLUE_DARKEST
48	#874CFF		135	76	255	VIOLET	260	LIGHT	VIOLET_LIGHT
49	#5400FF		84	0	255	VIOLET	260	FULL	VIOLET_FULL
50	#190064		25	0	100	VIOLET	260	DARK	VIOLET_DARK
51	#0F0030		15	0	48	VIOLET	260	DARKEST	VIOLET_DARKEST
52	#FF4CFF		255	76	255	MAGENTA	300	LIGHT	MAGENTA_LIGHT
53	#FF00FF		255	0	255	MAGENTA	300	FULL	MAGENTA_FULL

BYTE	HEX		R	G	B	FAMILY	HUE	SHADE	ENUM
54	#590059		89	0	89	MAGENTA	300	DARK	MAGENTA_DARK
55	#190019		25	0	25	MAGENTA	300	DARKEST	MAGENTA_DARKEST
56	#FF4C87		255	76	135	ROSE	340	LIGHT	ROSE_LIGHT
57	#FF0054		255	0	84	ROSE	340	FULL	ROSE_FULL
58	#59001D		89	0	29	ROSE	340	DARK	ROSE_DARK
59	#220013		34	0	19	ROSE	340	DARKEST	ROSE_DARKEST

ALIAS DEFAULT_ACTIVE = EMERALD_FULL (byte 25, firmware factory value)

3. Assorted (Selected)

17 constants promoted to the enum

BYTE	HEX		R	G	B	ENUM	GAP_DE
96	#FF7F00		255	127	0	S_TRUE_ORANGE	11.5
126	#B35F00		179	95	0	S_BROWN	16.6
105	#693C1C		105	60	28	S_BROWN_DARK	9.3
83	#402100		64	33	0	S_BROWN_DARKEST	8.7
61	#993500		153	53	0	S_RUST	11.9
109	#FFE126		255	225	38	S_GOLD	8.6
97	#B9B000		185	176	0	S_GOLD_DARK	18.3
74	#AFED06		175	237	6	S_CHARTREUSE	8.4
111	#67B50F		103	181	15	S_CHARTREUSE_DARK	16.6
62	#795100		121	81	0	S_OLIVE	14.6
125	#3F3100		63	49	0	S_OLIVE_DARK	9.9
112	#1E1E30		30	30	48	S_MIDNIGHT_BLUE	7.3
119	#E0FFFF		224	255	255	S_PALE_CYAN	11.4
115	#9A99FF		154	153	255	S_PERIWINKLE	12.3
93	#877FE9		135	127	233	S_PERIWINKLE_DARK	11.3
94	#D31DFF		211	29	255	S_PURPLE	7.5
82	#B21A7D		178	26	125	S_MAGENTA_ROSE	18.6

4. Assorted (Unselected)

45 assorted bytes not promoted, grouped by family

BYTE	HEX		R	G	B	NAME	GROUP
106	#A8000A		168	0	10	medium red	REDS_ORANGES
120	#A00000		160	0	0	medium red	REDS_ORANGES
121	#350000		53	0	0	very dark red	REDS_ORANGES
60	#FF1500		255	21	0	orange-red	REDS_ORANGES
107	#DE513D		222	81	61	light orange-red	REDS_ORANGES
84	#FF4A00		255	74	0	orange-red	REDS_ORANGES
108	#D86A1C		216	106	28	orange	REDS_ORANGES
127	#4B1502		75	21	2	very dark brown	BROWNS
99	#835D07		131	93	7	olive brown	BROWNS
100	#392B00		57	43	0	very dark olive	YELLOWS_OLIVES
63	#436400		67	100	0	dark olive	YELLOWS_OLIVES
113	#DCFF6B		220	255	107	light yellow-green	YELLOW-GREENS_LIMES
73	#BDFF2D		189	255	45	yellow-green	YELLOW-GREENS_LIMES
110	#9EE12F		158	225	47	lime	YELLOW-GREENS_LIMES
85	#88E106		136	225	6	lime	YELLOW-GREENS_LIMES
98	#90FF00		144	255	0	lime	YELLOW-GREENS_LIMES
86	#72FF15		114	255	21	lime	YELLOW-GREENS_LIMES
75	#64FF09		100	255	9	lime	YELLOW-GREENS_LIMES
122	#1AD000		26	208	0	green	GREENS
76	#108B00		16	139	0	dark green	GREENS
123	#074200		7	66	0	dark green	GREENS
88	#3BFF26		59	255	38	green	GREENS
101	#144C10		20	76	16	dark green	GREENS
64	#033900		3	57	0	very dark green	GREENS
89	#59FF71		89	255	113	light green	GREENS
114	#80FFBD		128	255	189	pale spring green	GREENS
77	#00FF87		0	255	135	spring green	GREENS
65	#005735		0	87	53	dark spring green	GREENS
102	#0D5038		13	80	56	dark spring green	GREENS
90	#38FFCC		56	255	204	turquoise	TURQUOISES_CYANS
68	#00454F		0	69	79	dark cyan	TURQUOISES_CYANS
66	#00547F		0	84	127	dark sky blue	BLUES
91	#5B8AFF		91	138	255	light blue	BLUES
92	#3151C6		49	81	198	medium blue	BLUES
79	#002AFF		0	42	255	blue	BLUES
104	#16205A		22	32	90	dark blue	BLUES
103	#15152A		21	21	42	very dark blue	BLUES

BYTE	HEX		R	G	B	NAME	GROUP
69	#2500CC		37	0	204	violet	VIOLETS_PURPLES
80	#3F00FF		63	0	255	violet	VIOLETS_PURPLES
116	#8E66FF		142	102	255	light violet	VIOLETS_PURPLES
81	#7A00FF		122	0	255	violet	VIOLETS_PURPLES
95	#FF005D		255	0	93	pink	MAGENTAS_PINKS
71	#202020		32	32	32	near-black grey	NEUTRALS
117	#404040		64	64	64	very dark grey	NEUTRALS
118	#757575		117	117	117	dark grey	NEUTRALS

5. Full palette

all 128 bytes, canonical machine-readable dump

BYTE	HEX		R	G	B	NAME	ENUM	DUP_OF
0	#000000		0	0	0	black	BLACK	
1	#1E1E1E		30	30	30	near-black grey	GREY_DARKEST	
2	#7F7F7F		127	127	127	mid grey	GREY_LIGHTEST	
3	#FFFFFF		255	255	255	white	WHITE	
4	#FF4C4C		255	76	76	light red	RED_LIGHT	
5	#FF0000		255	0	0	red	RED_FULL	
6	#7F0000		127	0	0	dark red	RED_DARK	
7	#1E0000		30	0	0	very dark red	RED_DARKEST	
8	#FFBD6C		255	189	108	light orange	ORANGE_LIGHT	
9	#FF5400		255	84	0	orange-red	ORANGE_FULL	
10	#591D00		89	29	0	dark brown	ORANGE_DARK	
11	#271B00		39	27	0	very dark olive	ORANGE_DARKEST	
12	#FFFF4C		255	255	76	light yellow-green	YELLOW_LIGHT	
13	#FFFF00		255	255	0	yellow-green	YELLOW_FULL	
14	#595900		89	89	0	dark olive	YELLOW_DARK	
15	#191900		25	25	0	very dark olive	YELLOW_DARKEST	
16	#88FF4C		136	255	76	light lime	LIME_LIGHT	
17	#54FF00		84	255	0	green	LIME_FULL	
18	#1D5900		29	89	0	dark green	LIME_DARK	
19	#142B00		20	43	0	very dark olive	LIME_DARKEST	
20	#4CFF4C		76	255	76	light green	GREEN_LIGHT	
21	#00FF00		0	255	0	green	GREEN_FULL	
22	#007F00		0	127	0	dark green	GREEN_DARK	
23	#001E00		0	30	0	very dark green	GREEN_DARKEST	
24	#4CFF5E		76	255	94	light green	EMERALD_LIGHT	
25	#00FF19		0	255	25	green	EMERALD_FULL	
26	#00590D		0	89	13	dark green	EMERALD_DARK	
27	#001902		0	25	2	very dark green	EMERALD_DARKEST	
28	#4CFF88		76	255	136	light spring green	JADE_LIGHT	
29	#28F656		40	246	86	green	JADE_FULL	
30	#0B681F		11	104	31	dark green	JADE_DARK	
31	#002413		0	36	19	very dark spring green	JADE_DARKEST	
32	#4CFFB7		76	255	183	light spring green	MINT_LIGHT	
33	#00FF99		0	255	153	spring green	MINT_FULL	
34	#005935		0	89	53	dark spring green	MINT_DARK	
35	#001912		0	25	18	very dark turquoise	MINT_DARKEST	
36	#4CC3FF		76	195	255	light sky blue	AZURE_LIGHT	
37	#00A9FF		0	169	255	sky blue	AZURE_FULL	
38	#004152		0	65	82	dark cyan	AZURE_DARK	
39	#001019		0	16	25	very dark sky blue	AZURE_DARKEST	
40	#4C88FF		76	136	255	light blue	COBALT_LIGHT	
41	#0055FF		0	85	255	blue	COBALT_FULL	
42	#001D59		0	29	89	dark blue	COBALT_DARK	
43	#000819		0	8	25	very dark blue	COBALT_DARKEST	
44	#4C4CFF		76	76	255	light blue	BLUE_LIGHT	
45	#0000FF		0	0	255	blue	BLUE_FULL	
46	#00007F		0	0	127	dark blue	BLUE_DARK	
47	#00001E		0	0	30	very dark blue	BLUE_DARKEST	
48	#874CFF		135	76	255	light violet	VIOLET_LIGHT	
49	#5400FF		84	0	255	violet	VIOLET_FULL	
50	#190064		25	0	100	dark violet	VIOLET_DARK	
51	#0F0030		15	0	48	very dark violet	VIOLET_DARKEST	
52	#FF4CFF		255	76	255	light magenta	MAGENTA_LIGHT	
53	#FF00FF		255	0	255	magenta	MAGENTA_FULL	
54	#590059		89	0	89	dark magenta	MAGENTA_DARK	
55	#190019		25	0	25	very dark magenta	MAGENTA_DARKEST	
56	#FF4C87		255	76	135	light pink	ROSE_LIGHT	

BYTE	HEX		R	G	B	NAME	ENUM	DUP_OF
57	#FF0054		255	0	84	pink	ROSE_FULL	
58	#59001D		89	0	29	dark pink	ROSE_DARK	
59	#220013		34	0	19	very dark pink	ROSE_DARKEST	
60	#FF1500		255	21	0	orange-red		
61	#993500		153	53	0	rust brown	S_RUST	
62	#795100		121	81	0	olive brown	S_OLIVE	
63	#436400		67	100	0	dark olive		
64	#033900		3	57	0	very dark green		
65	#005735		0	87	53	dark spring green		
66	#00547F		0	84	127	dark sky blue		
67	#0000FF		0	0	255	blue		45
68	#00454F		0	69	79	dark cyan		
69	#2500CC		37	0	204	violet		
70	#7F7F7F		127	127	127	mid grey		2
71	#202020		32	32	32	near-black grey	GREY_DARK	
72	#FF0000		255	0	0	red		5
73	#BDF52D		189	255	45	yellow-green		
74	#AFED06		175	237	6	yellow-green	S_CHARTREUSE	
75	#64FF09		100	255	9	lime		
76	#108B00		16	139	0	dark green		
77	#00FF87		0	255	135	spring green		
78	#00A9FF		0	169	255	sky blue		37
79	#002AFF		0	42	255	blue		
80	#3F00FF		63	0	255	violet		
81	#7A00FF		122	0	255	violet		
82	#B21A7D		178	26	125	medium magenta	S_MAGENTA_ROSE	
83	#402100		64	33	0	very dark brown	S_BROWN_DARKEST	
84	#FF4A00		255	74	0	orange-red		
85	#88E106		136	225	6	lime		
86	#72FF15		114	255	21	lime		
87	#00FF00		0	255	0	green		21
88	#3BFF26		59	255	38	green		
89	#59FF71		89	255	113	light green		
90	#38FFCC		56	255	204	turquoise		
91	#5B8AFF		91	138	255	light blue		
92	#3151C6		49	81	198	medium blue		
93	#877FE9		135	127	233	pale blue	S_PERIWINKLE_DARK	
94	#D31DFF		211	29	255	magenta	S_PURPLE	
95	#FF005D		255	0	93	pink		
96	#FF7F00		255	127	0	orange	S_TRUE_ORANGE	
97	#B9B000		185	176	0	mustard	S_GOLD_DARK	
98	#90FF00		144	255	0	lime		
99	#835D07		131	93	7	olive brown		
100	#392B00		57	43	0	very dark olive		
101	#144C10		20	76	16	dark green		
102	#0D5038		13	80	56	dark spring green		
103	#15152A		21	21	42	very dark blue		
104	#16205A		22	32	90	dark blue		
105	#693C1C		105	60	28	dark brown	S_BROWN_DARK	
106	#A8000A		168	0	10	medium red		
107	#DE513D		222	81	61	light orange-red		
108	#D86A1C		216	106	28	orange		
109	#FFE126		255	225	38	yellow	S_GOLD	
110	#9EE12F		158	225	47	lime		
111	#67B50F		103	181	15	olive	S_CHARTREUSE_DARK	
112	#1E1E30		30	30	48	very dark blue	S_MIDNIGHT_BLUE	
113	#DCFF6B		220	255	107	light yellow-green	DEFAULT_INACTIVE	
114	#80FFBD		128	255	189	pale spring green		
115	#9A99FF		154	153	255	pale blue	S_PERIWINKLE	
116	#8E66FF		142	102	255	light violet		
117	#404040		64	64	64	very dark grey	GREY_FULL	
118	#757575		117	117	117	dark grey	GREY_LIGHT	
119	#E0FFFF		224	255	255	very pale cyan	S_PALE_CYAN	
120	#A00000		160	0	0	medium red		
121	#350000		53	0	0	very dark red		
122	#1AD000		26	208	0	green		
123	#074200		7	66	0	dark green		
124	#B9B000		185	176	0	mustard		97
125	#3F3100		63	49	0	very dark olive	S_OLIVE_DARK	

BYTE	HEX		R	G	B	NAME	ENUM	DUP_OF
126	#B35F00		179	95	0	brown	S_BROWN	
127	#4B1502		75	21	2	very dark brown		